

How Quantum Is the Advantage? A Fair, Calibration- and Noise-Aware Benchmark and Attribution Audit of Quantum Machine Learning for Network Intrusion Detection

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Abstract

A quantum-kernel advantage that survives both false-discovery-rate and family-wise correction can be produced entirely by the tuning of the classical baseline. Benchmarking hybrid variational circuits and quantum-kernel SVMs against five tuned classical baselines on four network intrusion detection datasets under one leakage-controlled protocol, with a quantum-attribution audit, we find that tuned classical models match or exceed the quantum models on every dataset, yet two quantum results survive correction. The first, a quantum kernel out-ranking its classical surrogate, is produced by the classical baseline’s bandwidth selection: under NSL-KDD’s train-to-test shift, cross-validation is anticorrelated with the classical kernel’s test performance on all 11 training samples we drew, so spurious advantages of up to 0.17 ROC-AUC appear under narrow and wide grids alike, while at each kernel’s best setting the quantum kernel never leads by more

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than 0.01. Without shift, cross-validation ranks the grid correctly. Holding attack families out of training on three further datasets (26 runs) recreates the failure, and a label-free MMD between training and unlabelled test traffic tracks it (Spearman $\rho = -0.61$). The second, a four-qubit hybrid detecting more attacks than a tuned Random Forest at 1% false positives, is classical too: its trained circuit is exactly an 81-term trigonometric polynomial, and a classical tanh layer in the same pipeline matches it on every seed. On three IBM Heron processors the projected kernel stays within 0.01 ROC-AUC of simulation and the hybrid’s low-false-positive detection is unchanged. Significance is not attribution. Code, seeds, splits and raw device measurements are released.

Keywords: quantum machine learning, quantum kernels, dequantisation, distribution shift, network intrusion detection, hybrid quantum-classical neural networks, NISQ hardware, reproducible benchmark

1. Introduction

Network intrusion detection systems (NIDS) are a frontline defence against an expanding and increasingly automated threat landscape. Machine learning has become the dominant paradigm for flow- and packet-level intrusion detection, and the recent emergence of quantum machine learning (QML) has prompted a fast-growing body of work applying parameterised quantum circuits, quantum-kernel methods, and hybrid quantum-classical neural networks to the NIDS task. Headline results are striking: accuracies and F1 scores in the 97 to 99.9% range are reported across many datasets and quantum architectures.

These numbers invite a question the field has not answered convincingly: **how quantum is the reported advantage?** There are three reasons for scepticism, each grounded in rigorous recent work. First, the strongest quantum results are typically obtained on heavily dimensionality-reduced inputs, frequently one feature per qubit after principal component analysis (PCA), and on binary tasks evaluated under random cross-validation, a regime that systematically over-states real-world detection. Second, when honest, well-tuned classical baselines (Random Forests, gradient-boosted trees, modern deep networks) are placed on the *same* footing, they repeatedly match or beat the quantum models on tabular IDS data: the Meta-Quantum Ensemble study of Bhatnagar et al. [1], despite assembling the most rigorous evaluation in the field, explicitly concedes that classical ensembles remain superior. Third, and most pointedly, Bellante et al. [2] demonstrate that

the apparent advantage of PCA-based quantum IDS is *explainable by regularisation, not by quantum effects*. A reviewer of any new QML-IDS paper will, and should, raise this point.

We take this scepticism as the starting point rather than something to argue away. A paper whose central claim is “our quantum model achieves higher accuracy” is both scientifically fragile and strategically weak, because three Q1 papers already supply the counter-argument. The community’s actual deficit is not another accuracy record; it is a **fair, reproducible, and honest evaluation methodology**, together with a principled way to **attribute** any measured difference to its true cause. This paper supplies both.

We construct a single, leakage-controlled benchmarking protocol and apply it to the two quantum families that dominate the literature, namely hybrid variational quantum circuits (VQCs) and quantum-kernel support vector machines (QSVMs), against a slate of honestly tuned classical baselines, across four standard datasets. We honour each dataset’s intended evaluation semantics: for NSL-KDD we use the official `KDDTrain+`/`KDDTest+` split, whose test set deliberately over-represents attack behaviour under-seen in training, so that what we measure is generalisation to novel attacks rather than memorisation. We then go beyond benchmarking with a **quantum-attribution audit**: for every quantum model we construct a parameter-matched classical control sharing the identical classical front-end, and we run a regularisation sweep that tests the Bellante et al. hypothesis directly. Finally, because operational IDS must run at a low false-positive operating point and real quantum devices are noisy, we characterise how detection and calibration behave under simulated NISQ noise. Last, we dissect the audit’s two surviving positives, the kernel result at the level of the kernels themselves and the hybrid result through classical twins of its quantum layer, and show that both are classical: the first was produced by the tuning of the classical baseline under a distribution shift that standard cross-validation cannot see, the second by the input and the neural pipeline around the quantum layer. Both models reproduce their simulated results on IBM Heron processors.

1.1. Contributions

1. **A fair, reproducible, multi-dataset QML-IDS benchmark.** We evaluate hybrid VQCs and QSVMs against five honestly-tuned classical baselines (Random Forest, XGBoost, RBF-SVM, a tuned MLP, a 1D-CNN) across four standard datasets (NSL-KDD, UNSW-NB15, CICIDS2017, NF-ToN-IoT-v2) on the binary detection task, under one

protocol with strict leakage control, honouring official splits. All code, seeds, and splits are released.

2. **An equal-budget fairness design.** We report two explicitly-labelled feature views, a *full* view (classical models on all features) and a *same-budget* view (every model on the identical low-dimensional input the quantum models receive), eliminating the silent feature-budget asymmetry that confounds prior work.
3. **A quantum-attribution audit.** Parameter-matched classical controls and a regularisation sweep decompose any observed gain into its quantum and classical components, a direct constructive test of the Bellante et al. critique in the practical NISQ regime.
4. **A dissection of the audit’s two surviving positives.** Placing quantum and classical kernels on identical data with exact Gram matrices and equal tuning, we show that a quantum-kernel advantage surviving both FDR and Holm correction is produced by the classical baseline’s bandwidth selection, because cross-validation cannot see NSL-KDD’s train-to-test shift: on every one of 11 training samples it is anticorrelated with the classical kernel’s test performance, so widening the grid is no remedy, and the library-default SVM falls in the same trap. Controls on the test set alone show that cross-validation is reliable without the shift and apportion the artefact between novel attack types and a changed attack mix. For the audit’s other positive, a four-qubit hybrid’s advantage at 1% false positives, we build classical twins that replace only the quantum layer, show that the trained circuit is exactly an 81-term trigonometric polynomial, and find that a tanh layer of the same width reproduces the result, whose margin over the Random Forest lies entirely in novel attack types.
5. **An induced-shift protocol across three datasets and a label-free warning statistic.** Holding attack families out of training on UNSW-NB15, ToN-IoT and CICIDS2017 (26 runs) recreates the selection failure, including a spurious 0.17 quantum lead on CICIDS2017, while at its best setting the quantum kernel never leads by more than 0.003. The squared MMD between training and unlabelled test traffic tracks how badly cross-validation fails. Leave-attack-types-out cross-validation repairs selection on NSL-KDD but not under induced shift, so we recommend reporting kernel comparisons under several selection rules with the oracle bound. The geometric difference of Huang et al., on all four datasets, shows the IQP kernel to be reproducible by a classical kernel on its own circuit phases.

6. **Hardware validation of both models.** The projected kernel and the hybrid’s quantum layer run on three IBM Heron processors: the kernel’s classifier stays within 0.01 ROC-AUC of simulation, and the hybrid’s low-false-positive detection is unchanged on 6,000 test rows per device and indistinguishable from its classical twin’s on the same rows.
7. **Operationally meaningful evaluation under noise.** Beyond accuracy/F1 we report detection rate and false-positive rate together, low-FPR operating points, and calibration, and characterise degradation under depolarising, amplitude-damping, phase-damping, and readout-error channels.
8. **Statistical rigour and honest framing.** Every result is a mean over three to five seeds with confidence intervals, and we report the per-class seed budget explicitly (Section 5.2); quantum-vs-best-classical comparisons are paired over the seeds the two models share and use McNemar’s test, paired t /Wilcoxon tests, and a paired bootstrap with effect sizes. The benchmark’s value does not depend on the quantum models winning.

2. Related Work

We organise the related work into five strands and summarise representative studies in Table 1.

Foundational and breadth-oriented QML-IDS. Quantum intrusion detection was established by Kalinin and Krundyshev [3], who applied a quantum SVM and quantum convolutional network to a custom IoT stream dataset ($\sim 98\%$ accuracy); Gong et al. [4] produced the most prominent Q1 variational result (97.21% precision, VQCNN, KDD/NSL-KDD). The strongest *comparative* line is Abreu et al.: QML-IDS [5] and QuantumNetSec [6] span VQC, QSVM, and QCNN across three to four datasets, but with under-tuned classical baselines and no attribution of where gains originate.

Methods landscape. The dominant primitive is the parameterised quantum circuit used as a variational classifier or hybrid head; recurring concerns are trainability (barren plateaus) and encoding choice, with amplitude embedding reported as unstable relative to angle embedding [7]. Quantum-kernel SVMs [8] form the second family, scaling poorly in sample count, hence the reliance on subsampled, PCA-reduced data. Newer directions include quantum GNNs (Q-AGNN [9]) and meta-learning ensembles [1].

Rigorous and sceptical evaluations.. A small set of studies interrogates whether quantum helps. Bhatnagar et al.’s Meta-Quantum Ensemble [1] sets the field’s evaluation standard (AUPRC, TPR at low FPR, Brier, ECE, five noise channels) yet concludes classical ensembles remain superior. Bellante et al. [2], in *Computers & Security*, show the observed advantage is explainable by regularisation rather than quantum effects. These works define the bar we must clear: not a higher accuracy, but a fair and *attributable* comparison.

Quantum kernels under scrutiny.. Outside intrusion detection, a growing literature asks the same question of quantum kernels. Schnabel and Roth [10] benchmark fidelity and projected quantum kernels over 64 datasets and more than 20,000 models, and find that classical hyperparameters, notably regularisation and bandwidth, matter more than the number of qubits or layers, with no substantial difference between fidelity and projected kernels. Flórez-Ablan et al. [11] show that optimally bandwidth-tuned quantum kernels closely resemble RBF kernels and so hold no quantum advantage, and Shaydulin and Wild [12] establish the importance of the bandwidth itself. Kakavand et al. [13] find no significant advantage for quantum-kernel SVMs across nine tabular datasets, with hardware validation. Abdulsalam and Ahmad [14] compare fidelity quantum kernels with RBF, linear, polynomial and sigmoid kernels, in both classical and quantum SVMs, across eleven tabular classification and regression datasets, and Innan et al. [15] train the quantum kernel variationally to improve QSVM accuracy. Huang et al. [16] supply a label-free criterion, the geometric difference, for when a quantum kernel can outperform a classical one at all. To our knowledge none of these studies examines the shift between training and test data that defines NIDS evaluation. Within intrusion detection, Cotrupi and Callahan [17] run quantum kernels on hardware for four attack types, reporting an average accuracy of 85%. We extend this line to that shift, and show how it interacts with the tuning of the classical baseline. For variational models, Bowles et al. [18] find across 160 datasets that out-of-the-box classical models outperform quantum classifiers and that removing entanglement often costs nothing, and classical surrogates of trained variational circuits have been built from their Fourier structure [19, 20, 21]. We use such a surrogate as an attribution control: to test, inside the identical pipeline, whether a result that survived multiple-testing correction needs its quantum layer, and whether it survives on hardware.

Table 1: Representative QML-IDS studies and where this work differs (“sim” = simulator).

Paper	Method	Dataset(s)	Honest baseline?	Attribution?
Kalinin & Krundyshev [3]	QSVM, QCNN	Custom IoT stream	No	No
Gong et al. [4]	VQCNN	KDD / NSL-KDD	Partial	No
Abreu et al. [5, 6]	VQC / QSVM / QCNN	UNSW, CICIDS, CIC-IoT	Under-tuned	No
Bhatnagar et al. [1]	QSVM, QNN, RF	CICIDS, TON_IoT	Yes (cl. wins)	No
Bellante et al. [2]	FT PCA-QML	NIDS datasets	Yes	FT/PCA only
This work	VQC, QSVM, controls	4 datasets, official splits	Yes (5 tuned)	Yes (NISQ)

Datasets and evaluation practice.. Most studies use one or two datasets (NSL-KDD/KDD99 the dated-but-default choice); genuine multi-dataset and cross-dataset evaluation is rare, and imbalance-aware metrics, calibration, significance testing, and honest baselines are almost universally missing, with Bhatnagar et al. [1] the principal exception.

We differentiate on three axes. Against Abreu et al. we retain breadth but add tuned baselines, an equal-budget view, operating-point and calibration metrics, significance testing, and reproducibility. Against Bhatnagar et al. we add the causal *decomposition* their paper concedes is missing. Against Bellante et al., the sharpest existing critique of PCA-based quantum IDS, we extend the regularisation argument from their fault-tolerant, PCA-specific case study to the practical NISQ hybrid-VQC and QSVM models the community deploys, across a full multi-dataset benchmark, and we do so constructively: we specify exactly what evidence a genuine advantage would require and report whether, where, and to what extent it appears.

3. Threat Model and Background

3.1. Threat model

We consider flow- and connection-record-based detection: a model classifies each per-connection feature vector as benign or malicious. Three operational constraints shape the evaluation: a strong *cost asymmetry* (false positives flood analysts, so a detector must hold up at a low-FPR operating point); *distribution shift* between training and deployment, including

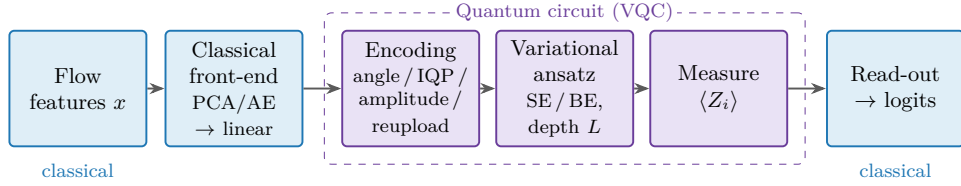


Figure 1: Hybrid quantum-classical classifier. A classical front-end compresses flow features to the per-qubit budget; a variational quantum circuit (encoding + ansatz + measurement) processes the encoded state; a classical read-out produces class logits. The quantum component is sandwiched between classical processing, which is the motivation for the attribution audit.

unseen attack signatures (why we honour the NSL-KDD official split); and *class imbalance*. Adaptive adversarial evasion is out of scope.

3.2. Background on quantum machine learning

A QML model operates on n qubits whose joint state is a unit vector in a 2^n -dimensional complex space; a noiseless simulation tracks this 2^n -entry vector, while a noisy simulation tracks a $2^n \times 2^n$ density matrix, whose quadratic cost caps our noisy runs at twelve qubits. Classical flow features must be *encoded* into the circuit (we ablate angle, amplitude, IQP/ZZ, and data-re-uploading encodings); a *variational* ansatz of trainable gates processes the state; and measured Pauli- Z expectations are mapped to class logits by a classical layer (Figure 1). An alternative is to use the circuit only as a fixed feature map and let a classical SVM learn the boundary: the *fidelity kernel* $K(x, x') = |\langle \phi(x') | \phi(x) \rangle|^2$ costs $O(N^2)$ circuit evaluations, while the *projected kernel* computes per-qubit expectations and applies a classical RBF, scaling as $O(N)$.

The link to our central question is direct: in every hybrid pipeline a *classical* front-end compresses high-dimensional records to the per-qubit budget and a *classical* read-out turns measurements into predictions, so any end-to-end accuracy blends a quantum contribution with substantial classical processing and implicit regularisation. Disentangling the two is exactly what the attribution audit (Section 4.4) is built to do.

4. Methodology

4.1. Benchmark protocol and leakage control

We evaluate every model under one protocol designed so no test-distribution information influences training, scaling, reduction, or model

selection. Where a dataset ships an official split we honour it: NSL-KDD uses the canonical **KDDTrain+**/**KDDTest+** split (17 attack signatures appear only in test, so we measure generalisation to novel attacks); UNSW-NB15 uses its official CSVs; CICIDS2017 and NF-ToN-IoT-v2 use a single seeded stratified 70/30 split. All preprocessing is fit on training data only; leaky identifier columns (IP/port four-tuples, flow IDs, timestamps, row indices) are dropped; model selection is by cross-validation *within the training set only*, and the test set is scored once per model per seed.

4.2. Two feature views (equal-budget fairness)

We report a **full view** (classical baselines on all encoded dimensions, the practical ceiling) and a **same-budget view** (every model on the identical PCA-reduced input, $n_{\text{features}} = n_{\text{qubits}} = 8$ in the main configuration). Because one-hot expansion produces many sparse columns, standard-scaling before PCA wastes the budget (eight components retain only $\sim 27\%$ of variance on NSL-KDD); min-max scaling before PCA lifts this to 0.841 (Table 2), so we min-max scale prior to reduction.

4.3. Models

The quantum side is a hybrid VQC family and two QSVMs through a single model factory: a parameter-matched classical MLP control, six hybrid VQCs over the encoding $\times$ ansatz grid (angle, IQP/ZZ, amplitude, re-uploading; strongly-entangling [SE] and basic-entangler [BE] ansätze), and two QSVMs (IQP fidelity-kernel, angle projected-kernel). The ZZ feature map is built from primitive gates for correct mini-batch broadcasting. The five classical baselines (Random Forest, XGBoost, RBF-SVM, tuned MLP, 1D-CNN) are each hyperparameter-searched (randomised search / grid, 5-fold stratified CV on the training set, optimising F1) and refit per seed.

4.4. The quantum-attribution audit

This is the methodological core and our constructive answer to Bellante et al. [2]. Figure 2 sketches the logic. The audit has three components: (i) *parameter-matched classical controls*: for each hybrid VQC, a classical MLP of matched parameter count sharing the identical front-end, and for each QSVM an RBF-SVM and a *random-feature* kernel approximation; if a matched control equals the quantum model, the “advantage” is not attributable to the quantum component; (ii) a *regularisation sweep* varying classical regularisation (read-out ridge, MLP weight decay, RF depth, SVM bandwidth) to test whether comparable regularisation closes any gap; and (iii) a *gain decomposition* reporting, per dataset and metric, the quantum-minus-control difference with significance.

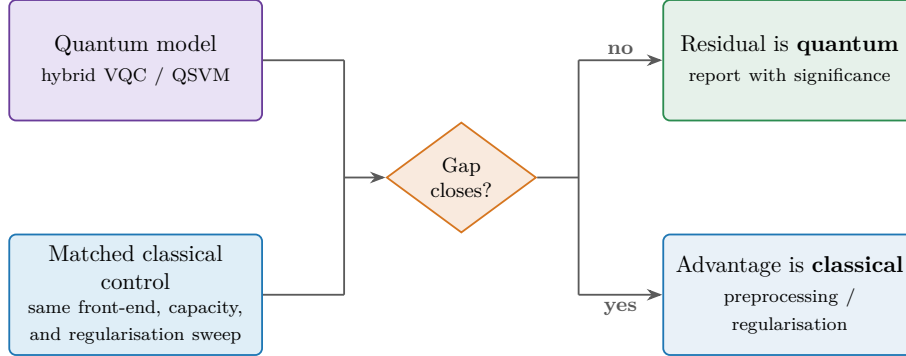


Figure 2: The attribution audit. Each quantum model is compared with a parameter-matched classical control sharing its front-end, plus a regularisation sweep. If a comparably-regularised classical model closes the gap, the apparent advantage is classical; any significant residual is attributed to the quantum circuit.

4.5. Kernel-level diagnostics

The audit above compares models. To explain *why* a quantum kernel does or does not help, we also analyse the kernels themselves. Every quantum kernel studied here is noiseless, so we simulate each feature-map state $|\phi(x)\rangle$ once and form every Gram matrix from the stored statevectors: the fidelity kernel as $|\Psi_A \Psi_B^\dagger|^2$, and the projected kernel from each qubit’s reduced density matrix. This costs $O(N)$ circuit simulations instead of $O(N^2)$, a shortcut available only in simulation, agrees with the circuits used for the QSVM to within 10^{-15} on every entry, and reproduces the published QSVM test metrics to four decimal places.

Fair surrogates.. We compare quantum and classical kernels on the identical first 2,000 NSL-KDD training rows, with exact Gram matrices and the same kernel SVM solver throughout. The bandwidth γ and $C \in \{0.1, 1, 10, 100\}$ are selected on the training rows only, with the same grid for quantum and classical kernels, so neither side receives more tuning than the other. We use two bandwidth grids, because the choice matters: a narrow one of seven values from 0.003 to 3, and a wide one of ten values from 0.001 to 30, both logarithmically spaced. Selection is either standard 5-fold stratified cross-validation or the shift-aware variant described in Section 7.2.

Shift controls.. Three controls change only the evaluation set of the NSL-KDD comparison, so that the training rows, the grid and the cross-validation stay fixed (Section 7.2): the official test set without the 17 attack types absent from the training file; that set importance-weighted so that each

attack type, and the benign class, carries its share of the training file; and 20,000 held-out rows of the training file, which share its distribution.

Induced shift.. To test whether shift causes the selection failure rather than merely accompanying it, we remove whole attack families from a seeded, stratified 2,000-row training sample while leaving them in the 20,000-row test sample, on the three datasets whose labels name the attack family (Section 7.4): every family in turn on UNSW-NB15 (nine), NF-ToN-IoT-v2 (nine) and CICIDS2017 (two), and two multi-family holdouts on UNSW-NB15 over three sampling seeds. For every run we also compute the unbiased squared maximum mean discrepancy [22] between the training sample and the test sample on the raw features (RBF kernel, median-heuristic bandwidth, at most 2,000 rows of each), which needs no test labels. Re-running a published landscape this way reproduces it to within 10^{-3} , the SVM solver’s tolerance under a different number of BLAS threads. As a sensitivity check on the shift-aware rule we also report a second fold construction, which splits benign rows evenly across folds and assigns attack types largest-first to the fold holding the fewest attack rows, so that every fold contains both classes.

Geometric difference and model complexity.. Following Huang et al. [16], for kernels normalised to $\text{Tr } K = N$ we compute

$$g(K_C \| K_Q) = \sqrt{\left\| \sqrt{K_Q} \sqrt{K_C} (K_C + \lambda I)^{-2} \sqrt{K_C} \sqrt{K_Q} \right\|_\infty},$$

which tends to $\sqrt{\left\| \sqrt{K_Q} K_C^{-1} \sqrt{K_Q} \right\|_\infty}$ as $\lambda \rightarrow 0$. A small g certifies that the classical kernel can learn any function the quantum kernel can learn from comparable data, for *any* labelling of the inputs, whereas a prediction advantage requires g to approach $\sqrt{N}$. We take the minimum of g over each classical kernel family and bandwidth, at $N = 800$ and $\lambda = 10^{-2}$. Rank-deficient kernels, such as the linear kernel on eight features, are excluded: their regularised inverse vanishes on the null space, which makes g spuriously small. We also report the label-dependent model complexity $s_K = \left\| \sqrt{K} (K + \lambda I)^{-1} y \right\|^2$ with $y \in \{\pm 1\}$, which tends to $y^\top K^{-1} y$ as $\lambda \rightarrow 0$.

Concentration.. We measure the mean and variance of the off-diagonal kernel entries as the qubit count grows from 2 to 12. Exponential concentration [23] would drive this variance to zero exponentially in the number of qubits.

Finite shots. Each Pauli expectation μ in the projected kernel is replaced by the mean of M outcomes in $\{\pm 1\}$ drawn with $P(+1) = (1 + \mu)/2$, which is what hardware returns from M shots in each of the three measurement bases, and the SVM is retrained on the resulting noisy features.

Hardware execution. Because the projected kernel needs only single-qubit expectations, the feature map can be run on a real device as one parameterised eight-qubit circuit with 24 observables, executed once per sample: linear in the number of samples, unlike the $O(N^2)$ fidelity kernel. We build the identical IQP circuit in Qiskit (its statevector expectations agree with the PennyLane feature map to 10^{-15}), transpile it to the device’s native gate set (depth 119, 28 two-qubit gates on the heavy-hex coupling map), and evaluate $\langle X_i \rangle, \langle Y_i \rangle, \langle Z_i \rangle$ with the Estimator primitive at 128 shots per measurement basis, with the primitive’s default readout-error mitigation and no other error mitigation. The 200 training rows are the first 200 of the NSL-KDD training file and the 500 test rows are a seeded, class-balanced sample of the official test set. All 700 samples are submitted as a single job. We compare the measured features and the resulting Gram matrix with their exact statevector values by Pearson correlation, and train and test the projected-kernel SVM on the measured features alone. As a control we first run the same pipeline on a noisy local simulation of the device built from its calibration data.

The same devices also run the variational model behind the audit’s other surviving positive, the four-qubit hybrid of Table 4. Its trained circuit (angle encoding, three strongly entangling layers) is rebuilt in Qiskit with the trained weights bound as constants and the four input angles as circuit parameters (statevector expectations agree with PennyLane to 10^{-15}), transpiled to the heavy-hex coupling map (depth 84 to 92, 30 two-qubit gates), and its four $\langle Z_i \rangle$ are measured at 128 shots on three disjoint, seeded, class-balanced subsets of 2,000 test rows, one job per subset and device, so that every device sees the same 6,000 rows, 3,000 of them benign. The measured expectations pass through the trained classical head unchanged, so only the quantum layer is replaced. The model is retrained from the released configuration and seed before the runs; the retrained model reproduces the published test scores to 2×10^{-8} . We compare it on the same rows with the exact simulation, the tuned Random Forest of Table 4 and the classical twins described next, by paired bootstrap.

Classical twins of the hybrid. To attribute the hybrid’s result we keep its pipeline intact (the four PCA features, the seeded 2,000-row training sub-

sample, the Linear(4, 4) and tanh input layer, the Linear(4, 2) output layer, Adam at learning rate 0.005, 20 epochs, batches of 64, class-weighted cross-entropy, and seeds 42 to 46) and replace only the quantum layer. With *R_Y* angle encoding, every $\langle Z_j \rangle$ of the circuit is a trigonometric polynomial in the four input angles a_i with frequencies in $\{-1, 0, 1\}$ [19], that is, a linear combination of the $3^4 = 81$ products $\prod_i f_i(a_i)$ with $f_i \in \{1, \cos a_i, \sin a_i\}$. The *Fourier twin* replaces the circuit by an unconstrained linear map on these 81 features, a strict superset of the circuit’s function class and a classical surrogate in the sense of Schreiber et al. [20], here trained from scratch rather than fitted to the trained circuit; the *MLP twin* by two tanh layers of width four (40 parameters against the circuit’s 36); and a third twin removes the middle layer altogether. Fitting the 81 coefficients to a trained circuit by least squares tests the expansion directly. To separate the model class from the input, we also train the Random Forest on the hybrid’s own input: its four features, on the full training file and on the same 2,000 rows.

Classical phase kernels.. Under angle encoding every qubit remains in a product state, so its Pauli features are exactly $(\sin x_i, 0, \cos x_i)$ and the “quantum” projected kernel is an RBF kernel on classical trigonometric features. We therefore also test RBF kernels on $\varphi(x) = (\sin x_i, \cos x_i)_i$, and on φ augmented with the phases that the ZZ feature map [24] writes into the state, namely sin and cos of $2x_i$ and of $2(\pi - x_i)(\pi - x_{i+1})$. This isolates whether an advantage requires the quantum state itself, or only the phases it encodes.

4.6. Class imbalance and noise

The NSL-KDD binary task is only mildly imbalanced, so we default to cost-sensitive learning (balanced class weights, `scale_pos_weight` for XGBoost, weighted cross-entropy) and treat SMOTE (training partition only) as an ablation. For noise robustness we sweep four channels (depolarising, amplitude damping, phase damping, and bit-flip readout) at strengths $p \in \{0, 0.001, 0.005, 0.01, 0.05\}$ on PennyLane’s `default.mixed` density-matrix backend, capped at twelve qubits.

5. Experimental Setup

5.1. Datasets

We evaluate on four standard NIDS datasets (Table 2). The **main benchmark is the binary task**. CICIDS2017 here is the three-day *MachineLearningCVE* subset (~ 1.04 M flows); NF-ToN-IoT-v2 is stratified-

Table 2: Datasets (binary task; PCA-EVR@8 = variance retained by 8 principal components).

Dataset	Split	Raw dim	Classes (bin / multi)	Train / test	EVR@8
NSL-KDD [25]	Official	122	2 / 40	125,973 / 22,544	0.841
UNSW-NB15 [26]	Official	194	2 / 10	175,341 / 82,332	0.870
CICIDS2017 [27]	Strat. 70/30	76	2 / 3	729,491 / 312,639	0.924
NF-ToN-IoT-v2 [28]	Strat. 70/30	39	2 / 10	140,000 / 60,000	0.974

downsampled to 200,000 rows (from 16.9M); both are standard, configurable scopes we disclose plainly.

5.2. Metrics and significance

We report threshold metrics (accuracy, precision, recall, F1, detection rate DR, false-positive rate FPR) and ranking/calibration metrics (ROC-AUC, AUPRC, TPR@0.1%FPR, TPR@1%FPR, Brier, ECE), matching and extending the Meta-Quantum-Ensemble suite [1]. We state the seed budget exactly, because it is uneven across model classes. The tuned classical baselines are run over five seeds {42, 43, 44, 45, 46} on NSL-KDD and over three seeds {42, 43, 44} on UNSW-NB15, CICIDS2017 and NF-ToN-IoT-v2; the quantum models and the attribution-audit controls over three seeds {42, 43, 44}; and the noise-robustness sweep over two seeds {42, 43}, with all RNGs pinned per seed and t -based 95% confidence intervals reported throughout. Every quantum-versus-classical comparison is therefore paired over the three seeds {42, 43, 44} that all models share, and uses McNemar’s test (shared test set), paired t /Wilcoxon tests with Cohen’s d (over seeds), and a paired bootstrap (2,000 resamples) for operating-point metrics. Because the attribution audit evaluates many variant×metric comparisons, we control the false-discovery rate across this family with the Benjamini-Hochberg (BH) procedure and report BH-adjusted q -values alongside raw p -values; we treat a result as robust only if it survives FDR control.

5.3. Backend efficiency and reproducibility

At QML-IDS qubit counts, quantum simulation (not GPU throughput) dominates runtime, and we report a **backend crossover**: the CPU `lightning.qubit` simulator is markedly faster than the GPU `lightning.gpu` simulator (Section 6.5). Each run writes a self-describing JSON record (config, per-seed metrics, tuned hyperparameters, reference-seed predictions,

environment snapshot with source checksums). All experiments run on a single laptop (NVIDIA RTX 3050 Ti, 4 GB, under WSL2); the main runs use eight qubits, with a ceiling of sixteen (noiseless) / twelve (noisy). Code, seeds, and splits are released.

6. Results

The overall result, qualified below: *honestly-tuned classical models match or exceed the quantum models on aggregate detection on every dataset, but a small hybrid VQC retains a statistically significant edge at the strict low-FPR operating point on the distribution-shifted NSL-KDD task.*

6.1. Headline benchmark

Table 3 reports F1 under both views. Under the *same-budget* view (the apples-to-apples comparison), the best classical baseline beats the leading 8-qubit quantum model on every dataset: NSL-KDD 0.782 vs. 0.730; UNSW-NB15 0.891 vs. 0.836; CICIDS2017 1.000 vs. 0.968; NF-ToN-IoT-v2 0.991 vs. 0.949. On NSL-KDD the strongest quantum *ablation* variants narrow this gap substantially, since the IQP-encoded QSVM and the 4-qubit hybrid reach F1 of 0.752 and 0.759 (Section 6.2), but tuned classical still leads on F1, and retains a clearer margin on AUPRC and ROC-AUC, so the aggregate ordering is unchanged. Two qualifications matter. First, **CICIDS2017 and NF-ToN-IoT-v2 are near-saturated** (every tuned classical model scores $F1 \geq 0.97$), so they have weak discriminative power and the quantum models’ high absolute scores there should be read as “this task is easy.” Second, the **CV-to-test gap** confirms why literature numbers are inflated: on NSL-KDD, models scoring $F1 \approx 0.98$ under within-training cross-validation collapse to $F1 \approx 0.78$ on the official test split, a consistent ≈ 0.20 drop across all models and both views. Figure 3 compares the best classical and best quantum F1 per dataset; the qubit-count ablation is shown later in Figure 4.

6.2. Attribution audit

The audit (Table 4) decomposes the difference on NSL-KDD, the discriminative dataset. A parameter-matched classical MLP sharing the quantum front-end attains AUPRC 0.944 and the tuned Random Forest attains F1 0.764, both matching or exceeding every hybrid VQC on aggregate metrics. For the twelve-qubit hybrid the classical controls win significantly ($F1 - 0.039$, $p = 0.021$; $AUPRC - 0.036$, $p = 0.035$; $TPR@1\%FPR - 0.121$, $p = 0.026$). **Once a classical model is granted the same front-end**

Table 3: Headline benchmark. F1 (mean over seeds): best classical per view vs. the leading 8-qubit quantum model per dataset (same-budget input). The final three columns give that quantum model’s AUPRC, ROC-AUC, and TPR@1%FPR. Stronger quantum ablation variants (e.g. the IQP-QSVM and the 4-qubit hybrid on NSL-KDD, F1 0.752 and 0.759) are analysed in Section 6.2.

Dataset	Classical F1		Quantum (8 qubits, same-budget)			
	full	same-budget	F1	AUPRC	ROC-AUC	TPR@1%FPR
NSL-KDD	0.803	0.782	0.730	0.911	0.865	0.411
UNSW-NB15	0.912	0.891	0.836	0.942	0.924	0.577
CICIDS2017	1.000	1.000	0.968	0.992	0.998	0.975
NF-ToN-IoT-v2	0.996	0.991	0.949	0.965	0.969	0.195

Best classical model per cell. Full view: SVM (NSL-KDD), MLP (UNSW), RF (CICIDS, NF-ToN-IoT-v2); same-budget: XGBoost (NSL-KDD), MLP (UNSW), RF (CICIDS, NF-ToN-IoT-v2).

and comparable capacity/regularisation, the aggregate “quantum advantage” disappears, a confirmation of the Bellante hypothesis on this dataset. Two exceptions survive Benjamini-Hochberg FDR control across the audit’s comparison family, however, and they are the most informative results in the audit. First, the quantum-kernel SVM significantly outperforms the random-feature (Nyström / RBF-sampler) kernel: on NSL-KDD the IQP-encoded QSVM beats the random-feature control on AUPRC ($\Delta = +0.075$, raw $p = 0.001$, BH $q = 0.011$), ROC-AUC ($\Delta = +0.134$, $p = 0.002$, $q = 0.018$), and TPR@1%FPR ($\Delta = +0.152$, $p = 0.017$, $q = 0.047$). Taken at face value this would isolate a genuinely quantum contribution, since the random-feature kernel is meant as the classical surrogate for the quantum feature map. Section 7 shows that it does not: the control differed from the QSVM in four further ways, and once those are removed the gap is set by the classical bandwidth search, while a classical kernel on the circuit’s own phases reproduces the quantum kernel. Second, the *four-qubit* hybrid beats the tuned Random Forest at the low-FPR operating point (TPR@1%FPR +0.050, $p = 0.005$, BH $q = 0.030$; Section 6.3); Section 7.7 shows that this one, too, is classical. The remaining quantum differences are either ties or significant *losses* to the classical controls (notably the twelve-qubit hybrid: F1 -0.039 , $p = 0.021$; AUPRC -0.036 , $p = 0.035$), consistent with overfitting at width rather than a quantum benefit. Within the audit, then, classical wins on aggregate and two FDR-robust quantum

Table 4: Quantum-attribution audit (NSL-KDD): leading quantum variants vs. best matched control. Δ = quantum – control (positive favours quantum, except ECE). q is the Benjamini-Hochberg FDR-adjusted value across the audit family; **bold** rows survive FDR control ($q < 0.05$). “Hyb.-Angle qn ” is the angle-encoded hybrid VQC on n qubits; rf-kern is the random-feature kernel and MLP-m the parameter-matched MLP.

Variant	Metric	Quantum	Best control	Δ	Q better?	p	q (BH)
QSVM-IQP	AUPRC	0.957	0.882 (rf-kern)	+0.075	yes	0.001	0.011
QSVM-IQP	ROC-AUC	0.945	0.811 (rf-kern)	+0.134	yes	0.002	0.018
QSVM-IQP	TPR@1%FPR	0.470	0.318 (rf-kern)	+0.152	yes	0.017	0.047
Hyb.-Angle q4	TPR@1%FPR	0.517	0.467 (RF)	+0.050	yes	0.005	0.030
Hyb.-Angle q4	TPR@0.1%FPR	0.407	0.272 (MLP-m)	+0.135	yes	0.428	0.62
Hyb.-Angle q4	F1	0.759	0.764 (RF)	−0.004	tie	0.776	0.84
Hyb.-Angle q12	F1	0.724	0.764 (RF)	−0.039	no	0.021	0.10
Hyb.-Angle q12	TPR@1%FPR	0.346	0.467 (RF)	−0.121	no	0.026	0.11

differences persist: the quantum kernel over the random-feature control, which Section 7 traces to the control, and the small hybrid at the strict operating point.

6.3. Operating-point and calibration analysis

The operating-point metrics hold one of the audit’s two FDR-robust quantum differences; the other, the quantum-kernel-vs-random-feature-kernel comparison, is analysed in Section 7. On NSL-KDD the four-qubit hybrid VQC achieves $\text{TPR@1\%FPR} = 0.517$, exceeding the best classical model (Random Forest, 0.467) by 0.050, statistically significant ($p = 0.005$, paired bootstrap). At 0.1%FPR it reaches 0.407 against 0.272, a difference that is not significant ($p = 0.43$). At the strict 1%-alarm budget a defender would actually impose, the small quantum model catches a larger fraction of attacks on novel-attack-heavy traffic. Calibration is comparable (ECE ≈ 0.18 to 0.22 for hybrids vs. 0.193 for Random Forest), the four-qubit hybrid marginally better. This is the pattern a genuine, if narrow, quantum signal would produce, though it survives FDR and not Holm correction (Section 6.5). Section 7.7 tests that reading and rejects it: classical twins that replace only the quantum layer reproduce the result.

6.4. Noise robustness

Sweeping four NISQ channels on the six-qubit hybrid shows **graceful degradation, and in the mild regime none at all**. Relative to the noiseless run (ROC-AUC 0.846, $\text{TPR@1\%FPR}$ 0.422, ECE 0.191), depolarising, amplitude-damping, and phase-damping noise at $p \in \{0.01, 0.05\}$ leave

Table 5: Backend efficiency. CPU (`lightning.qubit`) vs. GPU (`lightning.gpu`) training-step time at fixed work.

Qubits	CPU (s)	GPU (s)	GPU/CPU
4	1.12	8.89	8.0×
8	2.08	28.0	13.5×
12	3.94	54.3	13.8×

low-FPR detection essentially unchanged (0.421 to 0.430) and slightly *raise* ROC-AUC (to between 0.878 and 0.897), consistent with mild noise acting as a regulariser on the shifted task. Only bit-flip (readout) noise materially hurts (TPR@1%FPR 0.379 at $p = 0.05$). No channel causes catastrophic collapse: for this small-circuit regime, readout error is the one to mitigate.

6.5. Significance and efficiency

We control the false-discovery rate across the audit family (108 direction-aware comparisons on NSL-KDD) with Benjamini-Hochberg; 13 quantum-favouring and 27 classical-favouring differences survive at $q < 0.05$, so on balance the corrected evidence still favours classical on aggregate. On aggregate F1 the classical controls significantly beat quantum (twelve-qubit hybrid vs. RF, raw $p = 0.021$). The two quantum differences that survive FDR control are the quantum-kernel-vs-random-feature-kernel comparison (AUPRC $q = 0.011$, ROC-AUC $q = 0.018$) and the four-qubit hybrid at the low-FPR point ($q = 0.030$). We note honestly that the latter survives FDR (BH) but not the more conservative family-wise Holm correction across the full surface, under which only the quantum-kernel-vs-surrogate results remain significant. Statistical robustness is not attribution, however: Section 7 shows that the Holm-robust kernel result is produced by its classical control, and Section 7.7 that the FDR-robust hybrid result is reproduced by classical twins of its quantum layer. The backend-crossover finding (Table 5) is unambiguous: the **CPU simulator is 8 to 14× faster than the GPU** at QML-IDS qubit counts, the gap growing with width because the per-circuit work is too small to amortise GPU kernel-launch and transfer overhead. The projected quantum kernel ($O(N)$) is tractable at test scale whereas the fidelity kernel ($O(N^2)$) is not.

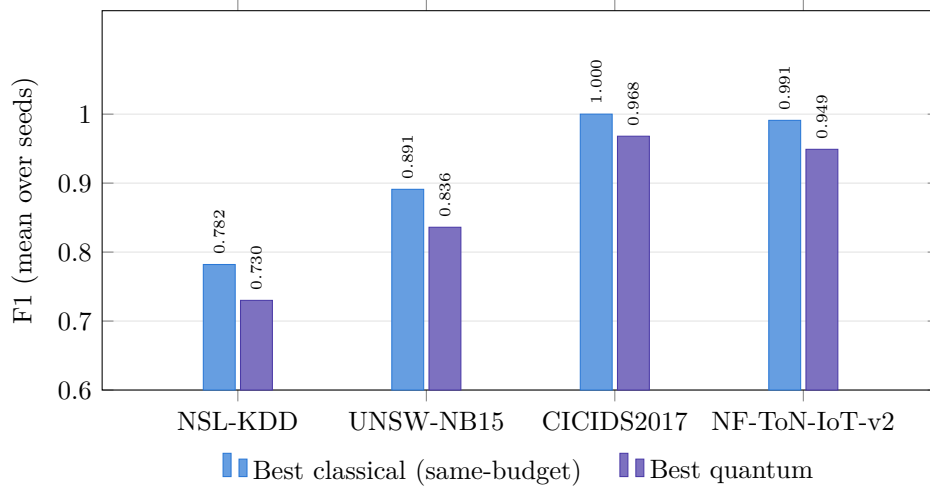


Figure 3: Best classical (same-budget) vs. best quantum F1 per dataset (mean over seeds). Tuned classical models lead on every dataset; the gap is largest on the discriminative datasets (NSL-KDD, UNSW-NB15) and smallest where the task is near-saturated (CICIDS2017, NF-ToN-IoT-v2, where every classical model exceeds 0.97). The vertical axis is truncated at 0.6.

7. Kernel- and Model-Level Analysis: How Quantum Are the Surviving Positives?

The audit’s most robust positive was the quantum-kernel SVM out-ranking its random-feature surrogate (Section 6.2), the one comparison to survive both FDR and family-wise Holm correction. That control, however, differed from the QSVM in four ways besides the feature map: it was trained on 1,200 rather than 2,000 rows, it approximated its kernel with 512 Nyström components, it used a linear rather than a kernel SVM, and its bandwidth $\gamma = 1$ was applied untuned to raw features on a different scale. This section removes each difference in turn, examines how the result depends on hyperparameter selection and on the shift between training and test data, tests it on a second official split and under shift induced on three datasets, and asks whether what remains requires the quantum state. It then asks the same question of the audit’s other surviving positive, the four-qubit hybrid, and finally runs both models on real devices. The diagnostics are those of Section 4.5.

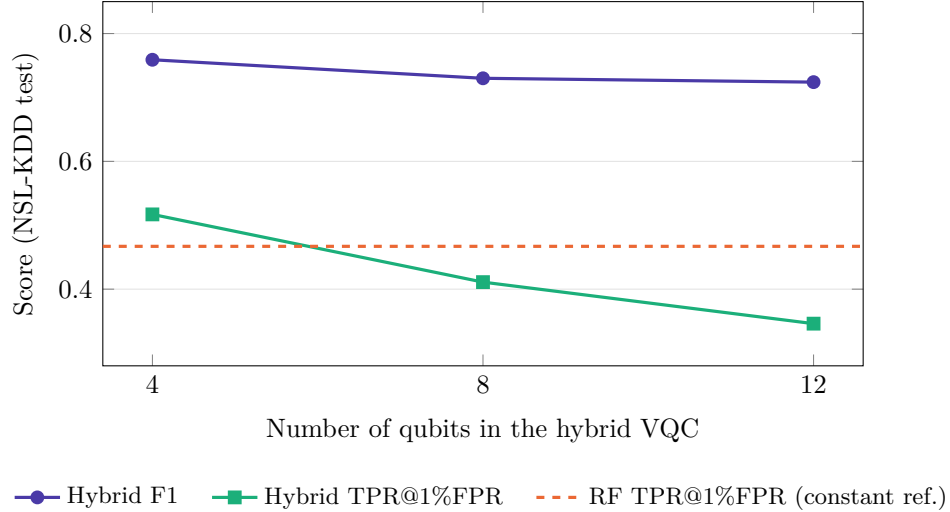


Figure 4: Qubit-count ablation of the hybrid VQC on NSL-KDD. Both F1 and the low-FPR detection rate *decline* with width, so more qubits do not help on the shifted task. Crucially, only the small four-qubit circuit exceeds the best classical baseline’s TPR@1%FPR (Random Forest, dashed reference at 0.467); the advantage vanishes by twelve qubits, consistent with an implicit-regularisation mechanism rather than raw expressivity.

7.1. The surrogate gap is set by the classical bandwidth search

Table 6 places every kernel on the identical 2,000 NSL-KDD training rows, with exact Gram matrices and one kernel SVM solver. Removing the first three confounds, with the bandwidth still fixed at $\gamma = 1$, narrows the published ROC-AUC gap from 0.127 to 0.085. Hyperparameter selection then decides the outcome. With the bandwidth searched over $\gamma \leq 3$, cross-validation picks $\gamma = 0.3$ for the raw-feature RBF kernel, whose test ROC-AUC falls to 0.783: an apparent quantum advantage of 0.167 (95% CI [0.161, 0.173]). With the same cross-validation over $\gamma \leq 30$ it picks $\gamma = 30$, and the RBF kernel reaches 0.940. The quantum kernel then leads by only 0.007 in ROC-AUC ([0.006, 0.009]) and *trails* at the operating point, where the classical kernel detects 0.455 of attacks at 1% false positives against 0.442 ($\Delta = -0.013$, $[-0.020, -0.005]$, $p = 0.002$). The advantage the audit reported is therefore not a property of the kernels. On these rows it appears or disappears with the width of the classical baseline’s search grid; on other training samples it appears under wide grids too, wherever cross-validation’s choice happens to fall (Section 7.2). It is a special case of the bandwidth sensitivity documented for quantum kernels by Shaydulin and Wild [12], here acting on the *classical* side of the comparison. Schnabel and Roth [10]

Table 6: Kernels on identical data (first 2,000 NSL-KDD training rows; official test set of 22,544 rows containing 17 attack types absent from training). Exact Gram matrices and one kernel SVM solver throughout. *Score* is the selection criterion’s own value: in-distribution 5-fold CV AUC, or leave-attack-types-out CV AUC for shift-aware rows. The phase kernel is an RBF kernel on $(\sin x_i, \cos x_i)$ plus the ZZ feature map’s phases, with no quantum state. Fixed rows use $\gamma = 1$, $C = 1$. Generated from the result files.

Kernel	Feat.	Selection	Score	Test AUC	AUPRC	TPR@1%FPR
<i>Quantum</i>						
IQP projected, published	24	fixed	n/a	0.945	0.957	0.470
IQP projected	24	CV, $\gamma \leq 3$	0.993	0.950	0.959	0.444
IQP projected	24	CV, $\gamma \leq 30$	0.994	0.948	0.957	0.442
IQP projected	24	shift-aware CV	0.941	0.945	0.962	0.499
IQP fidelity	256	CV, C only	0.992	0.947	0.955	0.458
<i>Classical, no quantum state</i>						
Random features, 1,200 rows, published	512	fixed	n/a	0.818	0.885	0.352
Random features, 2,000 rows	512	fixed	n/a	0.830	0.878	0.053
RBF, raw features	8	fixed	n/a	0.861	0.892	0.045
RBF, raw features	8	CV, $\gamma \leq 3$	0.993	0.783	0.864	0.078
RBF, raw features	8	CV, $\gamma \leq 30$	0.994	0.940	0.953	0.455
RBF, raw features	8	shift-aware CV	0.948	0.939	0.951	0.466
Trig $(\sin x_i, \cos x_i)$ only	16	CV, $\gamma \leq 3$	0.993	0.811	0.878	0.077
Phase kernel	46	CV, $\gamma \leq 3$	0.995	0.941	0.952	0.446
Phase kernel	46	CV, $\gamma \leq 30$	0.995	0.938	0.946	0.441
Phase kernel	46	shift-aware CV	0.951	0.947	0.959	0.469

likewise find bandwidth and regularisation to dominate quantum-kernel performance; what is new here is that a distribution shift makes the classical baseline’s choice of them unobservable to cross-validation.

7.2. Cross-validation cannot see the train-to-test shift

Figure 5 shows why the grid matters so much. Within the training distribution every kernel scores between 0.95 and 0.99 at every bandwidth for $C = 1$, and that is also what the kernels achieve on held-out rows of the training file: there, standard cross-validation ranks the 40 settings correctly for every kernel ($r = +0.94$ to $+0.99$; Table 7) and the narrow-grid artefact is absent ($\Delta\text{ROC-AUC} -0.002$). On the official test set the raw-feature RBF kernel is instead U-shaped in γ : strong at very wide and very narrow bandwidths (up to 0.945), and weakest at intermediate ones (0.740 at worst across the grid), which is exactly where a grid capped at $\gamma = 3$ leads cross-validation. Across the 40 settings its standard CV score is uncorrelated

with its test ROC-AUC ($r = -0.16$). The training rows and the cross-validation are the same in both cases; only the test distribution differs, and a criterion computed on the training distribution cannot anticipate how the shift reshapes the test landscape.

Table 7 separates the shift into its parts. Removing the 17 novel attack types from the test set (3,750 rows) leaves the blindness in place (RBF $r = -0.01$) and shrinks the narrow-grid artefact from 0.167 to 0.063 ROC-AUC. Reweighting the remaining rows to the training file’s attack-type mix shrinks it to 0.012, as the RBF landscape flattens (test ROC-AUC 0.97 to 0.99 at every setting, which also makes its r uninformative there). Of the aggregate artefact, the novel types thus account for about 60% and the changed attack mix for about 30%. At the operating point the decomposition differs: the RBF setting selected on the narrow grid detects 0.078 of attacks at 1% false positives on the official set, 0.090 without the novel types and 0.087 after reweighting, against 0.964 on held-out training rows. That collapse comes from drift within attack types the training file contains, not from the novel types.

Table 7: Decomposing NSL-KDD’s shift. The 2,000 training rows, the grid and the cross-validation are identical in every row; only the evaluation set changes. r is the correlation across the 40 grid settings between standard-CV ROC-AUC and test ROC-AUC. Under the narrow grid ($\gamma \leq 3$) and the wide grid ($\gamma \leq 30$), ΔAUC is the quantum projected kernel’s test ROC-AUC minus the raw-feature RBF kernel’s, each at its CV-selected setting; *RBF TPR* is that RBF setting’s TPR@1%FPR. The last column is the shift-aware (leave-attack-types-out) CV correlation for the RBF kernel. Reweighting gives each attack type present in the test set its share of the training file. Generated from the result files.

Test set	Rows	Standard CV r			Narrow grid		Wide	Shift-aware
		Quantum	Phase	RBF	ΔAUC	RBF TPR	ΔAUC	RBF r
Official test set	22,544	+0.83	+0.24	-0.16	+0.167	0.078	+0.007	+0.79
Without novel types	18,794	+0.76	+0.38	-0.01	+0.063	0.090	+0.008	+0.60
... reweighted mix	18,794	+0.94	+0.74	-0.36	+0.012	0.087	+0.001	+0.65
Held-out training rows	20,000	+0.99	+0.94	+0.95	-0.002	0.964	-0.002	+0.11

This motivates *shift-aware model selection*: leave-attack-types-out cross-validation, in which each attack type lives in exactly one fold, so every validation fold contains attack types its training fold never saw. It reproduces the condition of the official split using training data alone; attack types are assigned to folds by GroupKFold, and the three of five folds in which

both classes occur are used. Its score correlates with test ROC-AUC at $r = +0.79$ for the RBF kernel (+0.60 and +0.65 on the two reduced test sets of Table 7, but only +0.11 on held-out training rows, where there is no shift for it to anticipate), and it selects robust settings for every kernel family, with test ROC-AUC between 0.939 and 0.947 (Table 6). Under this rule the quantum projected kernel also reaches the highest operating-point detection we observe, $\text{TPR@1\%FPR} = 0.499$, above the RBF kernel’s 0.466 ($\Delta = +0.033$, $[0.021, 0.067]$, $p < 0.001$). We do not read this as a quantum advantage, because its sign depends on the selection rule: under standard CV the same comparison favours the classical kernel by 0.013. Having now evaluated three selection rules, reporting only the one on which quantum wins would be precisely the researcher degree of freedom this paper warns against.

Nor is the artefact a property of the published training rows. Table 8 repeats the comparison on ten further training samples, drawn at random and stratified by attack type, of 1,000, 2,000 and 4,000 rows. On all eleven samples the RBF kernel’s standard-CV score is anticorrelated with its test ROC-AUC ($r = -0.31$ to -0.07), so the setting that cross-validation selects is close to arbitrary with respect to the test set. The narrow grid yields a spurious quantum advantage of at least 0.05 ROC-AUC on six samples, and the wide grid on four, up to 0.142, with quantum leads at 1% false positives of up to 0.43: wherever the selection falls into the dip at $\gamma = 0.1$ to 1, whatever the width of the grid. The wide grid’s rescue in Section 7.1 is specific to the published rows, and more training data does not remove the effect (the two 4,000-row samples give 0.054 and 0.070 on the narrow grid). Two things hold on every sample. Shift-aware cross-validation, whose score correlates positively with test ROC-AUC on all of them ($r = +0.23$ to $+0.84$), keeps the gap at or below 0.066 (median 0.014), and at each kernel’s best setting in the grid the quantum kernel never leads by more than 0.010 (median 0.004). The comparison therefore has a stable answer, a tie, which standard selection under shift misses by up to 0.17 ROC-AUC in the quantum kernel’s favour.

The same dip catches the most common classical baseline in practice. Scikit-learn’s default RBF-SVM sets $\gamma = 1/(d \text{Var } X)$, which is 0.22 on the published training rows, inside the dip: with its default $C = 1$ it reaches test ROC-AUC 0.807 and detects 0.053 of attacks at 1% false positives, against 0.948 and 0.442 for the cross-validated quantum kernel. A comparison of a tuned quantum kernel with the library default would therefore report a quantum advantage of 0.14 ROC-AUC without any grid at all. On UNSW-NB15, where tuned classical kernels beat the quantum one (Section 7.3), the

Table 8: The surrogate artefact across training samples on NSL-KDD. Every row repeats the kernel comparison of Table 6 on a different training sample, the published first 2,000 rows or a seeded sample of the stated size stratified by attack type, and tests on the official test set. Each Δ is the quantum projected kernel’s test ROC-AUC (or TPR@1%FPR) minus the raw-feature RBF kernel’s, each at its own setting chosen by standard CV over the narrow grid ($\gamma \leq 3$), standard CV over the wide grid ($\gamma \leq 30$), shift-aware CV over the wide grid, or the best test ROC-AUC in the wide grid (oracle). r is the correlation across the 40 settings between the RBF kernel’s CV score and its test ROC-AUC, for standard and shift-aware CV. Generated from the result files.

Rows	Sample	Narrow, std. CV		Wide, std. CV		Wide, shift	Oracle	RBF r	
		Δ AUC	Δ TPR	Δ AUC	Δ AUC	Δ AUC	Δ AUC	Std.	Shift
1,000	random, seed 1	+0.008	−0.020	+0.005	+0.007	−0.006	−0.001	−0.24	+0.46
1,000	random, seed 2	+0.142	+0.181	+0.142	+0.005	−0.001	−0.001	−0.19	+0.23
1,000	random, seed 3	+0.016	−0.023	+0.010	+0.041	−0.001	−0.001	−0.25	+0.77
2,000	first rows (published)	+0.167	+0.367	+0.007	+0.006	+0.005	−0.001	−0.16	+0.79
2,000	random, seed 1	+0.010	+0.016	+0.010	+0.006	+0.001	−0.001	−0.31	+0.64
2,000	random, seed 2	+0.088	+0.322	+0.088	+0.014	+0.007	−0.001	−0.24	+0.56
2,000	random, seed 3	+0.020	−0.008	+0.008	+0.011	−0.000	−0.001	−0.25	+0.84
2,000	random, seed 4	+0.027	+0.016	+0.012	+0.066	+0.007	−0.001	−0.23	+0.54
2,000	random, seed 5	+0.087	+0.336	+0.087	+0.024	+0.010	−0.001	−0.18	+0.83
4,000	random, seed 1	+0.054	+0.429	+0.054	+0.015	+0.005	−0.001	−0.24	+0.68
4,000	random, seed 2	+0.070	+0.351	+0.011	+0.017	+0.004	−0.001	−0.07	+0.77

default still trails it (0.871 against 0.930).

7.3. A second official split

If the collapse of standard CV is caused by the shift, it should vanish on a split whose test set holds no novel attack types and little shift. UNSW-NB15 is the only other benchmark here with an official split, and every one of its nine attack categories occurs in both training and test. We repeat the protocol on a seeded, stratified sample of 2,000 training rows and 20,000 test rows; the training file is sorted by class, so its first rows are all benign and cannot be used. Table 9 confirms the prediction: on UNSW-NB15 standard CV tracks test performance for every kernel ($r = +0.82$ to $+0.83$). The result for the quantum kernel, however, reverses. Selected by the same rule, it is significantly worse than both classical kernels on every metric: the classical phase kernel of Section 7.5 reaches test ROC-AUC 0.956 against 0.930 ($\Delta = -0.026$, $[-0.028, -0.024]$) and detects 0.751 of attacks at 1% false positives against 0.457 ($\Delta = -0.294$, $[-0.318, -0.263]$), and even the raw-feature RBF kernel is ahead (Δ TPR = -0.057 , $[-0.087, -0.022]$, $p = 0.005$). The one property that had favoured the quantum kernel on NSL-KDD, fewer

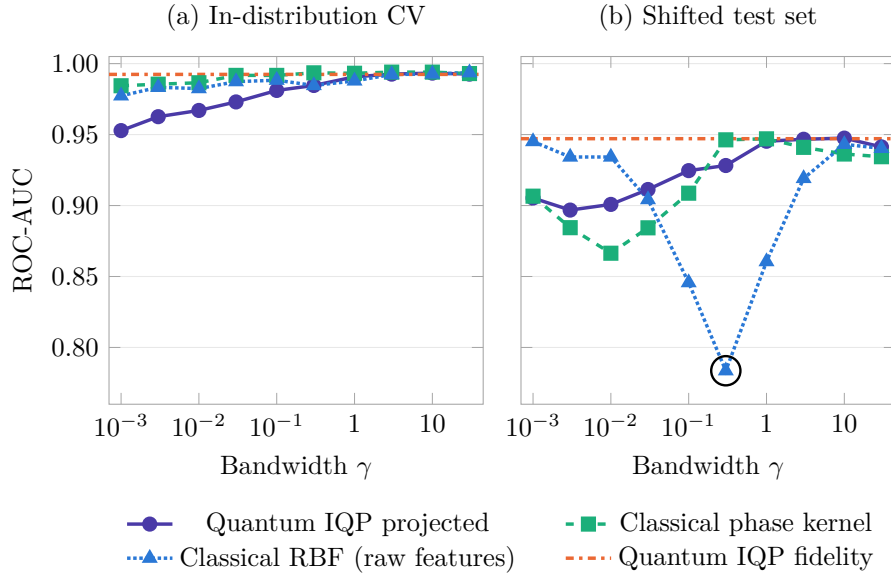


Figure 5: Kernel performance across the RBF bandwidth on NSL-KDD ($C = 1$, first 2,000 training rows). (a) Within the training distribution every kernel scores 0.95 to 0.99 at every bandwidth, as it does on held-out training rows (Table 7): without shift there are no bad settings to avoid. (b) On the official test set, shifted from the training distribution (Section 7.2), the raw-feature RBF kernel collapses at intermediate bandwidths while recovering at both extremes. Standard CV over a grid capped at $\gamma = 3$ selects $\gamma = 0.3$ ($C = 100$), test ROC-AUC 0.783, at the circled dip; with the grid extended to $\gamma = 30$ it selects $\gamma = 30$ ($C = 1$), test ROC-AUC 0.940. The classical phase kernel uses the ZZ feature map’s phases with no quantum state; the IQP fidelity kernel has no bandwidth and is drawn as a constant. On this dataset the quantum projected kernel never falls below 0.893 across the full grid, against 0.740 for the raw-feature RBF; on UNSW-NB15 the order reverses (Section 7.3).

poor hyperparameter settings (worst test ROC-AUC 0.893 against 0.740), does not survive either: on UNSW-NB15 its worst setting is the lowest of the three (0.697 against 0.787 and 0.789). We therefore do not claim it as a property of quantum kernels. Shift-aware CV on UNSW-NB15, whose official split has no novel attacks for it to anticipate, is examined with the induced-shift experiment in Section 7.4.

7.4. Inducing the shift on three datasets

The contrast between the two official splits is consistent with the shift explanation but does not test it, because the datasets differ in more than their splits. We therefore induce shift: whole attack families are removed

Table 9: Kernels on the two datasets with an official train/test split, each selected by standard 5-fold CV over the full grid ($\gamma \leq 30$, $C \leq 100$). *Worst* is the lowest test ROC-AUC over all 40 settings; r is the correlation, across those settings, between the CV score and the test ROC-AUC. NSL-KDD’s test set contains 17 attack types absent from training; UNSW-NB15’s contains none. Generated from the result files.

Dataset	Kernel	Test AUC	AUPRC	TPR@1%FPR	Worst	r
NSL-KDD	Quantum IQP projected	0.948	0.957	0.442	0.893	+0.83
	Phase kernel (classical)	0.938	0.946	0.441	0.818	+0.24
	RBF, raw features (classical)	0.940	0.953	0.455	0.740	−0.16
UNSW-NB15	Quantum IQP projected	0.930	0.943	0.457	0.697	+0.82
	Phase kernel (classical)	0.956	0.968	0.751	0.787	+0.83
	RBF, raw features (classical)	0.938	0.951	0.514	0.789	+0.83

from the 2,000-row training sample and left in the 20,000-row test sample, and the protocol of Table 9 is repeated (Section 4.5). On UNSW-NB15, ToN-IoT and CICIDS2017 each family is held out in turn (20 runs), and on UNSW-NB15 two multi-family holdouts are repeated over three sampling seeds: holdout A removes Exploits, Fuzzers and Reconnaissance (25% of the test rows), holdout B Generic, DoS, Analysis and Backdoor (29%). Table 10 summarises the 26 runs and Figure 6 shows each of them.

Shift is sufficient to recreate the failure, and its size tracks how badly. Without a holdout, standard CV tracks test performance on all three datasets (the smallest r over the three kernels is +0.82 to +0.99) and loses at most 0.019 ROC-AUC against the best setting in the grid. With families held out, the smallest r falls to +0.46 on ToN-IoT, +0.48 on UNSW-NB15 and −0.07 on CICIDS2017, and selection losses reach 0.061 for single UNSW-NB15 families, 0.112 under holdout B and 0.375 on CICIDS2017. The label-free squared MMD between the training sample and the unlabelled test sample orders the runs: it correlates with the smallest r at Spearman $\rho = -0.61$ (permutation $p = 0.001$) and with the largest loss at $\rho = +0.45$ ($p = 0.020$); its class-conditional version for the attack rows, which needs labels, reaches $\rho = -0.77$ and +0.66 ($p < 0.001$). It is a coarse signal rather than a test: CICIDS2017 with DDoS held out fails ($r = +0.03$) at a squared MMD of 0.006, while the holdout-A runs pass ($r \geq +0.83$) at 0.029 to 0.038.

What the failure does to a quantum-versus-classical comparison is what matters here. In 5 of the 26 runs selection reverses the sign of the quantum-minus-best-classical ROC-AUC gap. The largest case recreates the NSL-KDD artefact on a second dataset: with PortScan held out of CICIDS2017’s training sample, the CV-selected quantum kernel leads the best CV-selected classical kernel by 0.174 ROC-AUC, while at each kernel’s best setting in the grid it trails by 0.028. Across all 26 runs the quantum kernel never leads by more than 0.003 at the oracle settings. Nor is the outcome stable across training samples: under holdout B the kernel that selection penalises most changes with the sampling seed (the RBF kernel loses 0.108, 0.023 and 0.101 over the three seeds, the phase kernel 0.016, 0.084 and 0.112), and the quantum kernel’s lead at 1% false positives on the first seed (0.301 against 0.263 for the RBF kernel) does not recur on the other two (0.440 against 0.664, and 0.425 against 0.508).

Shift-aware selection does not repair the induced failures. Under holdout B (first seed), leave-attack-types-out CV, with a single usable fold, correlates with the RBF kernel’s test ROC-AUC at $r = +0.43$ and chooses a worse setting than standard CV (0.751 against 0.794): its folds can only hold out attack types the training sample contains, and a shift towards those is not the shift towards Generic and DoS that the test set presents. On NSL-KDD the seventeen novel types are variants within the same four attack families as the seen ones, which is why holding out types imitates the official split there. The result is not an artefact of the fold construction either. GroupKFold, used throughout, leaves three usable folds on NSL-KDD and one on UNSW-NB15; a construction that makes all five folds usable gives $r = +0.54$ instead of $+0.79$ for the RBF kernel on NSL-KDD and $r = -0.38$ under holdout B, and on the unshifted UNSW-NB15 split it selects an over-smooth RBF setting (test ROC-AUC 0.859 against 0.938 under standard CV). Shift-aware selection is a choice of target rather than a free improvement: it buys robustness to the kinds of novelty the training data can imitate, at a cost when there is no novelty, and it cannot anticipate novelty of a kind the training data has never seen. A kernel comparison under distribution shift should therefore be reported under more than one selection rule, with the oracle bound alongside and a label-free shift statistic such as the MMD, so that a reader can see how much of a claimed difference is selection.

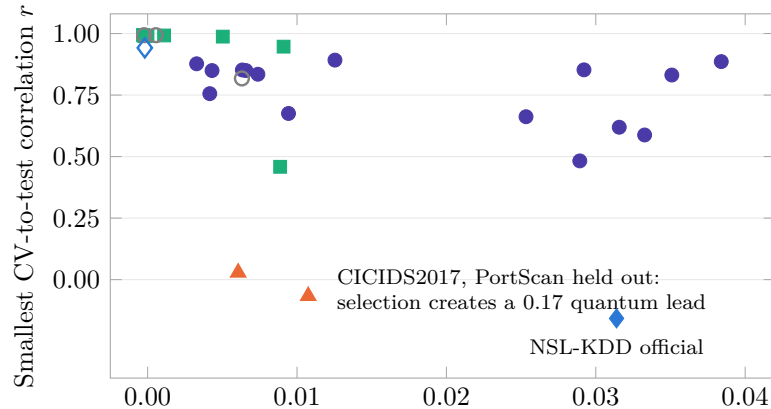
7.5. *Geometry: the IQP kernel is close to its own classical phases*

Table 11 gives the geometric difference on all four datasets, which requires no labels and no hyperparameter selection. The angle-encoded fidelity kernel is reproduced by an RBF kernel to $g = 1.06$ to 1.07 on every

Table 10: Inducing shift on three datasets. Whole attack families are removed from the 2,000-row training sample and kept in the 20,000-row test sample; every kernel is selected by standard 5-fold CV over the full grid. *Min r* is, per run, the smallest correlation over the three kernels between CV score and test ROC-AUC across the 40 settings (median and range over the group’s runs); *Loss* is the largest gap between a kernel’s best test ROC-AUC in the grid (oracle) and its CV-selected one. *Flips* counts runs in which the sign of the quantum-minus-best-classical ROC-AUC gap differs between the CV-selected and the oracle settings; *Sel. error* is the largest difference between those two gaps, and *Q lead* the largest quantum lead at the oracle settings. Across the 26 runs with a holdout, the label-free squared MMD between the training sample and the unlabelled test sample correlates with the smallest *r* at Spearman $\rho = -0.61$ (permutation $p = 0.001$) and with the largest loss at $\rho = +0.45$ ($p = 0.020$). Generated from the result files.

Runs	<i>n</i>	Min <i>r</i>		Loss		Flips	Sel. error	Q lead
		Median	Range	Median	Max			
No holdout	3	+0.99	[+0.82, +0.99]	0.003	0.019	0	−0.017	−0.001
UNSW-NB15, single family	9	+0.85	[+0.59, +0.89]	0.014	0.061	2	−0.021	+0.000
UNSW-NB15, A and B, 3 seeds	6	+0.75	[+0.48, +0.89]	0.048	0.112	0	+0.075	−0.007
ToN-IoT, single family	9	+0.99	[+0.46, +0.99]	0.005	0.019	2	−0.008	+0.001
CICIDS2017, single family	2	−0.02	[−0.07, +0.03]	0.230	0.375	1	+0.202	+0.003

dataset, so by the bound of Huang et al. [16] it can hold no prediction advantage over that classical kernel for any labelling. This is expected: under angle encoding every qubit remains in a product state, and the kernel factorises into a product of cosines. The IQP kernels, which use entangling ZZ gates, are genuinely far from standard kernels: against every RBF bandwidth and against plain trigonometric features, g ranges from 4.9 to 12.2. Flórez-Ablan et al. [11] show that optimally rescaling the encoding drives quantum kernels towards RBF kernels; our IQP kernels use the unscaled encoding of the original pipeline, and consistently with that result they sit far from RBF. The same IQP kernels, however, lie within $g = 1.4$ to 2.4 of a purely classical kernel built on the phases the ZZ feature map writes [24], on all four datasets, and their label-dependent model complexities agree: on NSL-KDD the quantum projected kernel has $s_K = 235$ and the classical phase kernel $s_K = 236$, against 481 for the best raw-feature RBF. The empirical results are consistent with this geometry. On NSL-KDD the phase kernel matches the quantum kernel at the operating point under a narrow grid (0.446 against 0.444, $p = 0.89$) and under standard CV over the full grid (0.441 against 0.442, $p = 0.90$), and on UNSW-NB15 it outperforms it.



Label-free shift: squared MMD, training sample vs. unlabelled test sample

- UNSW-NB15 holdout ■ ToN-IoT holdout ▲ CICIDS2017 holdout
- No holdout ◆ NSL-KDD official split ◆ NSL-KDD i.i.d. control

Figure 6: Label-free shift predicts when cross-validation stops ranking kernel settings. Each point is one kernel landscape (three kernels, 40 settings each): attack families held out of the 2,000-row training sample on UNSW-NB15, ToN-IoT and CICIDS2017 (filled), the same datasets without a holdout (open), and NSL-KDD’s official split and its i.i.d. control (diamonds). The horizontal axis needs no test labels. Across the 26 holdout runs, Spearman $\rho = -0.61$ (permutation $p = 0.001$). Generated from the result files.

Whatever the IQP kernel contributes lives in the phases of its circuit, not in the quantum state, and a classical kernel that uses those phases obtains it at a small fraction of the cost.

7.6. Concentration and finite shots

Kernel values concentrate as width grows, but mildly at these sizes. From 2 to 12 qubits the mean off-diagonal IQP fidelity falls from 0.32 to 0.047, yet its variance falls only from 0.099 to 0.032 and flattens between 8 and 10 qubits, far from the exponential regime of Thanasilp et al. [23]. The kernels’ test ROC-AUC peaks at 8 to 10 qubits (0.944 to 0.946 for the projected kernel) and dips only to 0.927 at 12. Concentration therefore does not explain the decline of the variational hybrid with width (Figure 4), which points instead to the trainability and capacity of the variational model.

The projected kernel is robust to measurement noise (Table 12). With 16 shots in each of three measurement bases, 48 shots per sample in total, test ROC-AUC falls from 0.945 to 0.941, and AUPRC and TPR@1%FPR do not move beyond their run-to-run spread; by 64 shots the result is indis-

Table 11: Geometric difference $g(K_C\|K_Q)$, minimised over classical bandwidth ($N = 800$, $\lambda = 10^{-2}$, $\sqrt{N} \approx 28.3$). A small g means the classical kernel can learn anything the quantum kernel can. The last column gives the angle-encoded fidelity kernel against its best RBF. Rank-deficient kernels are excluded. Generated from the result files.

Dataset	IQP kernel	Classical family, vs IQP kernel			Angle fidelity
		RBF raw	Trig	Phase kernel	vs RBF
NSL-KDD	projected	8.41	8.33	2.27	1.07
	fidelity	4.91	4.88	1.77	
UNSW-NB15	projected	12.21	12.10	2.44	1.07
	fidelity	9.52	9.34	1.50	
CICIDS2017	projected	8.65	8.48	1.95	1.06
	fidelity	7.04	6.86	1.54	
NF-ToN-IoT-v2	projected	9.25	9.07	2.23	1.07
	fidelity	7.84	7.64	1.39	

Table 12: Finite-shot robustness of the published IQP projected QSVM on NSL-KDD. M shots per basis in each of the X, Y and Z settings; mean and standard deviation over 5 sampling seeds. Generated from the result files.

Shots / basis	Shots / sample	ROC-AUC	AUPRC	TPR@1%FPR
16	48	0.941 ± 0.002	0.956 ± 0.002	0.486 ± 0.011
64	192	0.945 ± 0.002	0.960 ± 0.001	0.489 ± 0.009
256	768	0.943 ± 0.004	0.957 ± 0.002	0.473 ± 0.003
1,024	3,072	0.943 ± 0.002	0.956 ± 0.001	0.469 ± 0.002
4,096	12,288	0.947 ± 0.001	0.957 ± 0.001	0.470 ± 0.001
exact	∞	0.945	0.957	0.470

tinguishable from the exact kernel. This follows from the construction: the projected kernel needs only single-qubit expectations, each estimated with variance at most $1/M$, which the RBF then smooths, whereas a fidelity kernel must resolve small global overlaps. The study models sampling noise only; Section 7.8 tests the same kernel, with its gate and readout errors, on real devices.

7.7. The variational positive is classical

The audit’s second surviving positive is the four-qubit hybrid’s higher detection than the tuned Random Forest at 1% false positives on NSL-KDD (Section 6.3). The Random Forest was the strongest classical model on that metric, but it is not a control for the hybrid: it is a different model class,

it sees eight features rather than four, and it trains on the whole training file rather than on the hybrid’s 2,000 rows. Table 13 holds all of that fixed except the quantum layer (Section 4.5), and adds the Random Forest on the hybrid’s own input.

The trained circuit is classical in a precise sense. Fitting the 81-term trigonometric expansion to the circuit trained with seed 42 reproduces its four expectations to 2×10^{-15} on held-out angles, and the classical model built from those coefficients reproduces the hybrid’s test scores to 4×10^{-7} , the precision of single-precision arithmetic, with identical metrics. Entanglement is not idle, since without the 16 four-way products the error rises to 0.044, but every term it contributes can be written down. Trained from scratch in the same pipeline, the classical twins also match what the hybrid learns where it matters. Over five seeds the hybrid detects 0.515 ± 0.007 of attacks at 1% false positives, the tanh twin 0.511 ± 0.006 , the Fourier twin 0.506 ± 0.016 and the network with no middle layer at all 0.506 ± 0.010 , against 0.467 for the published Random Forest. The hybrid never differs significantly from the tanh twin (0 of 5 seeds) and leads the other two twins on 2 of 5 seeds each, by at most 0.021. On ROC-AUC the network without a middle layer is ahead of the hybrid (0.914 against 0.899). This matches the benchmark of Bowles et al., in which removing entanglement from quantum classifiers often cost nothing [18]; here removing the circuit altogether costs 0.009 at the operating point and gains 0.015 ROC-AUC.

The margin over the published forest therefore has classical parts, and the Random Forest controls size them. About half is the input: the same forest on the hybrid’s four features detects 0.491, and the hybrid’s lead over it shrinks to 0.023 to 0.030, significant on one of three seeds. Most of the rest is the model class, which a tanh layer supplies (0.511), and the quantum layer adds 0.004 over that layer, within noise. The split by attack type shows where the neural models differ. On attacks of types present in training every model detects 0.60 to 0.65 at 1% false positives; on the 3,750 attacks of the novel types the neural models detect 0.24 to 0.25 and the forest at most 0.15, whatever input it is given. The one variational result that survived the audit is the behaviour of a small, class-weighted neural network on four principal components, which extrapolates to novel attacks at a strict alarm budget better than a tree ensemble, and the quantum layer adds nothing to it that a tanh layer of the same width does not.

7.8. On real hardware

Table 14 reports the projected kernel executed on three IBM Heron processors, `ibm_fez`, `ibm_kingston` and `ibm_marrakesh`, on the same 200

Table 13: Attributing the four-qubit hybrid’s low-false-positive result on NSL-KDD. The upper block shares the hybrid’s pipeline (the same four PCA features, seeded 2,000-row training subsample, Linear(4, 4) and tanh input layer, Linear(4, 2) output layer, optimiser, loss, epochs and seeds), and only the middle layer differs; *Params* counts its trainable parameters. The lower block is the tuned random forest (RF), as published and on the hybrid’s input. Values are mean $\pm$ standard deviation over five seeds for the upper block and three for the lower, on the full official test set (22,544 rows). TPR@1%FPR is also split into attacks of types present in the training file (9,083 rows) and of the novel types (3,750 rows), at the same threshold. *Sig.* counts the seeds on which the hybrid’s TPR@1%FPR differs from that model’s at $p < 0.05$ (paired bootstrap). Generated from the result files.

Model	Params	ROC-AUC	TPR@1%FPR			Sig.
			All	Seen types	Novel types	
Hybrid QNN (quantum circuit)	36	0.899 $\pm$ 0.006	0.515 $\pm$ 0.007	0.624 $\pm$ 0.004	0.251 $\pm$ 0.029	
Fourier twin	328	0.865 $\pm$ 0.039	0.506 $\pm$ 0.016	0.616 $\pm$ 0.006	0.239 $\pm$ 0.048	2/5
Tanh MLP twin	40	0.885 $\pm$ 0.019	0.511 $\pm$ 0.006	0.620 $\pm$ 0.006	0.247 $\pm$ 0.019	0/5
No middle layer	0	0.914 $\pm$ 0.003	0.506 $\pm$ 0.010	0.612 $\pm$ 0.008	0.249 $\pm$ 0.018	2/5
RF, 8 features (published)		0.937 $\pm$ 0.003	0.467 $\pm$ 0.004	0.627 $\pm$ 0.007	0.079 $\pm$ 0.002	3/3
RF, 8 features, 2,000 rows		0.944 $\pm$ 0.006	0.481 $\pm$ 0.007	0.640 $\pm$ 0.007	0.097 $\pm$ 0.010	3/3
RF, 4 features		0.918 $\pm$ 0.003	0.491 $\pm$ 0.003	0.646 $\pm$ 0.003	0.115 $\pm$ 0.002	1/3
RF, 4 features, 2,000 rows		0.938 $\pm$ 0.011	0.470 $\pm$ 0.004	0.601 $\pm$ 0.006	0.152 $\pm$ 0.023	3/3

training and 500 test samples, each as a single job of 89 to 91 QPU-seconds (Section 4.5). The measured Pauli features correlate with their exact values at $r = 0.92$ to 0.97 (RMSE 0.10 to 0.15), and the Gram matrices built from them at $r = 0.97$ to 0.98 ; on `ibm_fez` the Gram correlation of 0.977 matches the $r \geq 0.976$ that Kakavand et al. [13] report for the same device. The classifier trained and tested on measured features alone reaches ROC-AUC 0.935, 0.935 and 0.942 against 0.944 for the exact kernel: a deficit of 0.009 on `ibm_fez` (95% CI [0.000, 0.018], $p = 0.049$) and on `ibm_kingston` ([0.002, 0.018], $p = 0.017$), and none on `ibm_marrakesh` ([−0.002, 0.007], $p = 0.35$). AUPRC shows a significant deficit only on `ibm_kingston` (0.011, $p = 0.002$), and at the 1% false-positive operating point no device differs from the exact kernel (measured 0.404 to 0.408 against 0.396, $p \geq 0.62$). The mechanism anticipated in Section 7.6 holds on hardware: per-feature noise well above the shot-noise floor is largely absorbed by the RBF over 24 features.

Two caveats apply. The devices were noisier at the feature level than the calibration-based noise model of the dry run predicted (test RMSE 0.15

Table 14: Hardware validation of the IQP projected kernel on NSL-KDD (200 training and 500 test samples, the latter class-balanced). Agreement columns are Pearson correlations of the source’s Pauli features and of its Gram matrix (off-diagonal) with the exact statevector values. Metric columns are for the projected-kernel SVM trained and tested on that source’s own features ($\gamma = 1$, $C = 1$). Generated from the result files.

Feature source	Agreement with exact		Projected-kernel SVM		
	Feat. r	Gram r	ROC-AUC	AUPRC	TPR@1%FPR
Exact statevector	1.000	1.000	0.944	0.941	0.396
Noisy simulation (FakeFez, 256 shots)	0.984	0.990	0.941	0.938	0.420
ibm_fez (128 shots)	0.920	0.977	0.935	0.935	0.404
ibm_kingston (128 shots)	0.941	0.971	0.935	0.930	0.404
ibm_marrakesh (128 shots)	0.966	0.984	0.942	0.939	0.408

on `ibm_fez` against 0.07 in simulation), so noise models of this kind should not be taken as substitutes for device runs. And hardware validity is not advantage: the same measured features are, by Section 7.5, reproducible by a classical kernel on the circuit’s phases. What the device runs establish is that the object we analysed is realisable at current noise levels, at a cost of about 0.13 QPU-seconds per sample.

The device runs also test the hybrid. Its four expectations, measured at 128 shots on the same 6,000 class-balanced test rows on each device (three disjoint subsets of 2,000, one job each), correlate with their exact values at $r = 0.972$ to 0.977 (Table 15). ROC-AUC falls from 0.893 to between 0.869 and 0.876 ($p < 0.001$ on every device), of which shot noise alone accounts for 0.012 in simulation of the first subset, while detection at 1% false positives is unchanged: 0.505 to 0.517 against 0.513 exact ($p \geq 0.76$). The margin over the Random Forest on these rows, +0.132 to +0.144 ($p = 0.003$ to 0.012), therefore survives real device noise on three processors. It is also the margin the classical twins show on the same rows (+0.136 to +0.153), and the hardware hybrid does not differ from the tanh twin on any device (-0.009 to +0.004, $p \geq 0.47$), so the result of Section 7.7 holds on hardware as well. Pooling was necessary: on a single 2,000-row subset the 1% threshold rests on ten benign scores, and the forest’s detection varied from 0.304 to 0.517 across the three subsets, its deficit to the hybrid from 0.21 to zero. The nine

jobs used 86 to 87 QPU-seconds each, about 0.04 QPU-seconds per sample.

Table 15: The 4-qubit hybrid on hardware: NSL-KDD, 6,000 class-balanced test rows per device (three disjoint subsets of 2,000, one job each, 86 to 87 QPU-seconds per job). Only the four $\langle Z_i \rangle$ that feed the trained classical head are measured; r is their correlation with the exact values. The lower block gives, on the same rows, the tuned random forest and the classical twins of Table 13 (same seed as the model run on hardware). TPR@1%FPR is also split into attacks of types seen in training and of the novel types. The last two columns are paired-bootstrap differences in TPR@1%FPR against the random forest and against the tanh MLP twin (two-sided p in brackets). Generated from the result files.

Source	r	ROC-AUC	TPR@1%FPR			Δ TPR@1%FPR vs	
			All	Seen	Novel	Random forest	MLP twin
Exact simulation	1.000	0.893	0.513	0.603	0.300	+0.140 [0.007]	+0.000 [0.92]
ibm_fez	0.972	0.870	0.517	0.607	0.304	+0.144 [0.003]	+0.004 [1.00]
ibm_kingston	0.973	0.876	0.511	0.603	0.293	+0.138 [0.008]	−0.002 [0.56]
ibm_marrakesh	0.977	0.869	0.505	0.603	0.271	+0.132 [0.012]	−0.009 [0.47]
Random forest, 8 features		0.933	0.373	0.505	0.060		
No middle layer		0.916	0.509	0.618	0.251	+0.136 [0.008]	
Tanh MLP twin		0.867	0.513	0.626	0.245	+0.140 [0.002]	
Fourier twin		0.880	0.526	0.617	0.310	+0.153 [0.001]	

7.9. Summary

No quantum advantage survives attribution in this benchmark. The audit’s two positives, each significant after correction, were produced classically: the quantum kernel’s by its surrogate’s confounds and then by bandwidth selection that the distribution shift leaves blind, and the hybrid’s by the input and the small neural pipeline around its quantum layer, which a tanh layer of the same width reproduces. Three findings of general use come out of the exercise. First, the selection failure is caused by train-to-test shift: it is absent without shift, present on every NSL-KDD training sample we drew, recreated on three datasets by holding attack families out of training, including a spurious 0.17 quantum lead, and tracked, coarsely, by a label-free MMD; no training-only rule repairs it in general, so comparisons under shift should be reported under several selection rules with the oracle bound. Second, both quantum models are classically reproducible for identifiable reasons: the IQP kernel from its own circuit phases on every dataset, by the geometric difference, and the hybrid’s trained circuit as an exact 81-term trigonometric polynomial. Third, both run on current hardware without changing these conclusions: on three IBM Heron processors the projected

kernel stays within 0.01 ROC-AUC of simulation, and the hybrid’s low-false-positive detection is unchanged on 6,000 rows per device, as is its tie with its classical twin.

8. Discussion

How quantum is the advantage? On this benchmark and in the practical NISQ regime, none that survives attribution. The audit settles the aggregate question: once a classical model is granted the same front-end and comparable regularisation, the aggregate advantage of the hybrid VQC and the QSVM vanishes, and tuned classical baselines significantly outperform the quantum models on F1 and AUPRC. We thus confirm the Bellante et al. [2] hypothesis and extend it from their fault-tolerant, PCA-specific case study to the deployed NISQ models, and we corroborate Bhatnagar et al.’s concession [1] while supplying the decomposition their work lacked.

The kernel-level analysis settles the exception we had expected to be the most robust. The quantum kernel’s advantage over its classical surrogate survived both FDR and family-wise correction, yet it was not quantum: it came from four confounds in the surrogate and then from the classical bandwidth selection, which the shift leaves blind (Section 7). This is a cautionary result for the field, not only for this paper. Statistical robustness does not establish attribution, and under the distribution shift that makes NIDS evaluation meaningful, the classical baseline’s hyperparameter search can manufacture or erase an apparent quantum advantage. A QMLIDS claim evaluated under distribution shift should report how its classical baselines were tuned, over what range, and by which selection rule; should report the comparison under more than one rule with the oracle bound, since no training-only rule, shift-aware selection included, proved reliable under every shift we induced; and can report a label-free shift statistic between training and test traffic as a warning.

The narrower positive the audit left, the four-qubit hybrid’s higher detection than the tuned Random Forest at the 1%-FPR operating point on NSL-KDD ($p = 0.005$, BH $q = 0.030$), is classical as well (Section 7.7). It was strongest for the smallest circuit and vanished at twelve qubits, which we had read as implicit regularisation rather than expressivity; the classical twins confirm that reading and remove the quantum layer from it. About half of the margin is the input, since a Random Forest on the hybrid’s four features closes it by half, and nearly all of the rest is the neural model class: a tanh layer of the same width, or no middle layer at all, in the same pipeline reproduces it, and the gain lies in novel attack types, to which small

neural networks on four principal components extrapolate better than tree ensembles. It reproduces on three IBM Heron processors, so it is not an artefact of noiseless simulation, but neither is it quantum. For variational models the lesson is that the control attribution requires is a classical twin that replaces only the quantum layer; a strong classical baseline on different inputs, however well tuned, cannot supply it.

What remains classical. For a practitioner choosing a detector today, the honest recommendation is a tuned Random Forest or XGBoost: they win or tie on every aggregate metric, on every dataset, at a fraction of the cost and with no quantum hardware. Four methodological findings generalise. The ≈ 0.20 F1 CV-to-test gap is a caution for the whole QML-IDS literature; leave-attack-types-out cross-validation addresses the model-selection failure behind it on NSL-KDD, the induced-shift experiment shows where such rules stop working, and a label-free MMD between training and test traffic flags the risk; classical twins that replace only the quantum layer are the control a variational claim needs; and the backend crossover (the GPU is an order of magnitude *slower* at these qubit counts) saves researchers real time.

9. Limitations and Threats to Validity

Hardware. The projected quantum kernel and the four-qubit hybrid’s quantum layer were executed on three IBM Heron processors (Section 7.8); every other quantum result, including the wider hybrids and the noise sweep, is simulated, and the noise sweep models NISQ effects rather than measuring them. The device runs used 200 training and 500 test samples for the kernel and 6,000 test samples per device for the hybrid, 128 shots per basis, and only the primitive’s default readout mitigation; they realise inference, not training, on hardware, and the fidelity kernel, whose cost is quadratic in the sample count, was not run. Pooled over three subsets, the hybrid’s 1% threshold rests on 30 benign scores per device, and its confidence intervals remain wide (for the margin over the forest, $[0.015, 0.220]$ on `ibm_marrakesh`). **Qubit budget.** The 4 GB GPU caps width at roughly sixteen qubits (noiseless) / twelve (noisy, $(2^n)^2$ memory); the main configuration uses eight, so we cannot probe the larger-width regime. **Dimensionality reduction** discards variance (eight PCs retain $\sim 84\%$ on NSL-KDD), so any advantage is conditioned on information-reduced input. **Dataset age/composition.** NSL-KDD is dated (mitigated by three further datasets); the IoT coverage is a single NetFlow dataset; CICIDS2017 and NF-ToN-IoT-v2 use standard subsets, configurable to full corpora. **Trainability** (barren plateaus, gradient variance) is not characterised directly, and QSVMs are trained on

subsampled data ($O(N^2)$ fidelity kernel). **Statistical power.** Our seed budget is small and uneven (five seeds for the classical baselines on NSL-KDD, three elsewhere and for all quantum runs, two for the noise sweep), so paired tests over seeds have low power: the two-sided Wilcoxon floor at three paired seeds is $p = 0.25$, which is why the operating-point claims rest on the paired bootstrap over the fixed test set rather than on seed-level tests. The crossover ratio is hardware/library-dependent. **Kernel and model analysis.** The kernel-level analysis uses eight qubits, the angle and IQP feature maps, noiseless statevectors, the two datasets with official splits and induced shifts on three datasets, each with 2,000 training rows; the geometric difference covers all four datasets, at $N = 800$. Shift-aware cross-validation requires attack-type labels that production data may not provide; the MMD statistic is a coarse, correlational signal from 26 runs, not a calibrated test; and the composition reweighting of Table 7 treats each attack type’s test rows as representative of that type. The classical twins attribute the four-qubit, angle-encoded hybrid only: wider or data-re-uploading circuits have larger trigonometric spectra, and the exact expansion grows as 3^n . The NSL-KDD QSVM, as in the original pipeline, trains on the first 2,000 rows of the official training file, so that model has a single realisation rather than three independent seeds; Table 8 repeats the kernel comparison on random training samples of 1,000 to 4,000 rows. **Scope.** Binary detection under benign distribution shift and imbalance; adversarial evasion and the unsupervised/online/federated settings are out of scope.

10. Conclusion

We set out to replace the field’s recurring “our quantum model reaches 99%” claim with a fair, reproducible, attribution-aware yardstick. Across four datasets under one leakage-controlled protocol, well-tuned classical models match or exceed the quantum models on aggregate detection everywhere, and the audit attributes this to classical preprocessing and regularisation rather than quantum effects, confirming and generalising the Bellante et al. critique. Neither of the audit’s two positives survived attribution. The quantum kernel that out-ranked its classical surrogate under both false-discovery-rate and family-wise correction was produced by the surrogate’s confounds and by the classical bandwidth selection, under a train-to-test shift that standard cross-validation cannot see and that, induced on three further datasets, manufactures quantum leads of the same size; a classical kernel on the circuit’s own phases reproduces the quantum kernel on all four datasets. The four-qubit hybrid’s advantage at a strict alarm budget belongs

to its input and its neural pipeline: its trained circuit is an exact 81-term trigonometric polynomial, and a tanh layer in its place reproduces the result. Both models run on IBM Heron processors with their simulated results intact, so hardware realisability is not attribution either. The lesson reaches beyond this paper: in QML benchmarks evaluated under distribution shift, significance is not attribution, and the tuning of the classical baseline, like the classical twin of the quantum layer, must be reported as carefully as the quantum model itself. We contribute the benchmark and harness, the kernel-level diagnostics, test-set shift controls and an induced-shift protocol with a label-free warning statistic, classical twins for variational models, and a shift-aware selection rule together with its limits, and release them so that QML-IDS claims can be stated, and contested, on common, honest ground.

Reproducibility. Code, configurations, fixed seeds, split scripts, and per-run provenance are released at <https://github.com/Orqly-AI/quantum-ids-benchmark>; the preprocessed datasets will be released on request.

CRedit authorship contribution statement

Syeda Anshrah Gillani and Mirza Samad Ahmed Baig are the core contributors and contributed equally to all aspects of the work. **Syeda Anshrah Gillani:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing (original draft), Writing (review & editing), Visualization, Supervision. **Mirza Samad Ahmed Baig:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing (original draft), Writing (review & editing), Visualization, Supervision. **Shahid Munir Shah:** Methodology, Validation, Writing (review & editing). **Abdul Akbar Khan:** Validation, Writing (review & editing). **Muhammad Omer Khan:** Validation, Writing (review & editing). **Asher Ali:** Resources, Writing (review & editing). **Hamzah Siddiqui:** Investigation, Validation, Writing (review & editing).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All datasets used are public benchmarks (NSL-KDD, UNSW-NB15, CICIDS2017, NF-ToN-IoT-v2) obtainable from their original providers. The code, configurations, fixed seeds, the download and preprocessing scripts that reconstruct our exact splits, and the per-run result records are released at <https://github.com/Orqly-AI/quantum-ids-benchmark>. We do not redistribute the source corpora; the preprocessed datasets will be released on request.

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Declaration of generative AI and AI-assisted technologies

During the preparation of this work the authors used AI-assisted tooling for coding assistance. The authors reviewed and edited all content and take full responsibility for the content of the publication.

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